MATHEMATICS

FUNCTIONS, VARIABLES, SEQUENCES AND SUMS

Function notation

Example contexts
I: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ **II**: $g : [0, \infty) \to \mathbb{R}$, $g(x) = \sqrt{x}$

Symbol	Meaning	Examples
f	Name of a function	I: $f(x) = x^2$ II: $g(x) = \sqrt{x}$
$f(x)$	Value of f at input x	I: $f(3) = 3^2 = 9$ II: $g(4) = \sqrt{4} = 2$
$f : A \to B$	Function from set A to set B	I: $f : \mathbb{R} \to \mathbb{R}$ II: $g : [0, \infty) \to \mathbb{R}$
$x \overset{f}{\mapsto} f(x)$	Input-to-output mapping	I: $x \overset{f}{\mapsto} x^2$; $3 \overset{f}{\mapsto} 9$ II: $x \overset{g}{\mapsto} \sqrt{x}$; $4 \overset{g}{\mapsto} 2$
$\mathrm{dom}(f)$	Domain of a function	I: $\mathrm{dom}(f) = \mathbb{R}$ II: $\mathrm{dom}(g) = [0, \infty)$

Letters and Greek symbols

Symbol	Name / description
$a, b, c, \ldots$	Common lowercase Latin letters
$A, B, C, \ldots$	Common uppercase Latin letters
x, y, z	Other common Latin letters
$\alpha,\ \beta,\ \gamma$	Alpha; beta; gamma
$\delta,\ \Delta$	Delta; uppercase delta
$\theta,\ \Theta$	Theta; uppercase theta
$\lambda,\ \Lambda$	Lambda; uppercase lambda
$\mu,\ \sigma$	Mu; sigma
$\omega,\ \Omega$	Omega; uppercase omega

Variable Examples

A variable is a symbol that can represent different values. Its allowed values and meaning depend on the context.

Possible values in separate cases

Each choice below is a separate case.

Ordinary numbers: x can be -2, 0, or 3.5.

Integers: n can be $1, 2, 3, \ldots$ when counting objects.

Angles: θ can be $30°$, $60°$, or $90°$.

Sets: A can be $\{1, 2\}$ or $\{0, 3, 5\}$.

Greek letters as numbers: λ can be -1, 0, or 2.

Letters do not have permanently assigned types. Uppercase letters often represent sets, and Greek letters often represent angles or numbers, but the context determines their meaning.

Lists, sequences and subscripts

A **sequence** is an ordered list of terms. A **series** is a sum of terms. Write sequences with parentheses and sets with braces.

Terms have positions

The terms $a_1, a_2, a_3, \ldots, a_n$ occupy positions $1, 2, 3, \ldots, n$. The notation a_i means **the term at position** i. The subscript i is an **index**, not multiplication or an exponent.

a_1 = first term
a_2 = second term
a_n = term at position n

One finite sequence, three forms

An explicit list, its indexed terms and an indexed family can describe the same sequence:

$$(2, 4, 6, 8) = (a_1, a_2, a_3, a_4) = (a_i)_{i=1}^{4}, \quad a_i = 2i.$$

In general, $(a_i)_{i=1}^{n} = (a_1, a_2, \ldots, a_n)$.

A rule generates the terms

For $a_i = 2i$, with $i = 1, 2, 3, 4$:
$a_1 = 2(1) = 2$; $a_2 = 2(2) = 4$;
$a_3 = 2(3) = 6$; $a_4 = 2(4) = 8$.
The resulting sequence is $(2, 4, 6, 8)$.

Order and repetitions matter

$(2, 4) \neq (4, 2)$ and $(2, 2, 4) \neq (2, 4)$.
For sets, order and repetition do not matter:
$\{2, 4\} = \{4, 2\} = \{2, 2, 4\}$.

From a sequence to its sum

The sum of the terms is $a_1 + a_2 + \cdots + a_n$. Summation notation writes this more compactly.

Summation notation

Read the parts

$$\sum_{i=1}^{n} a_i$$

$\sum$: summation symbol
i: index; 1: lower bound
n: upper bound
a_i: expression being summed

The index runs through consecutive integers from the lower bound to the upper bound, **inclusive**.

$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$.

Sum the sequence above: $a_i = 2i$

$$\sum_{i=1}^{4} a_i = \sum_{i=1}^{4} 2i = 2 + 4 + 6 + 8 = 20.$$

Other summands and bounds

Constant: $\sum_{i=2}^{4} 3 = 3 + 3 + 3 = 9.$

Index expression: $\sum_{i=1}^{2} (i + 3) = 4 + 5 = 9.$

Squares: $\sum_{k=1}^{3} k^2 = 1 + 4 + 9 = 14.$

The index name can change

$\sum_{i=1}^{4} i = \sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10.$
Use the new index name consistently in the bounds and summand.